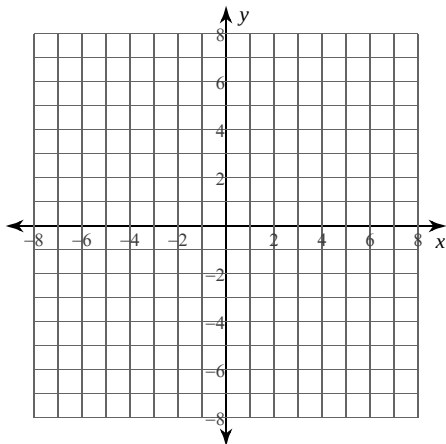
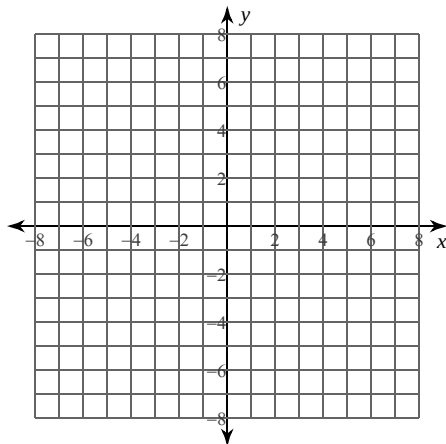


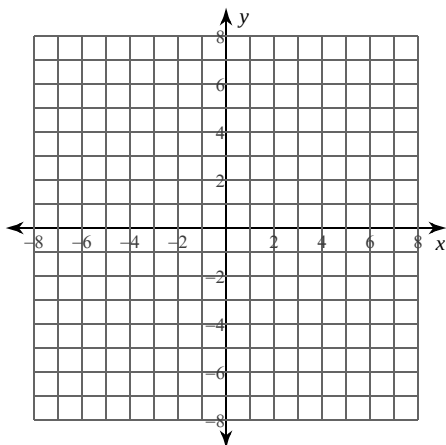
$$5) f(x) = \begin{cases} \frac{1}{x-4}, & x \leq 4 \\ -4, & x > 4 \end{cases}$$



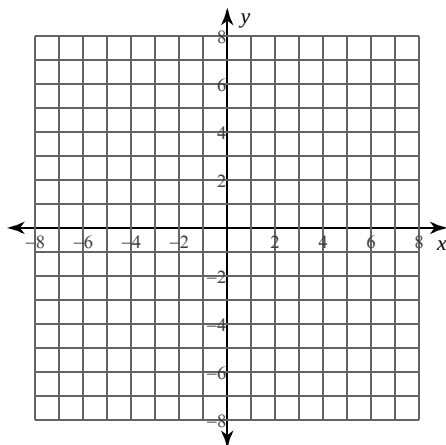
$$6) h(x) = \begin{cases} 2^x, & x < -3 \\ \frac{3}{x}, & x \geq -3 \end{cases}$$



$$7) w(x) = \begin{cases} \frac{|x|}{2}, & x \leq -4 \\ \sqrt{-x}, & -4 < x < 2 \\ |x-2|, & x \geq 2 \end{cases}$$



$$8) w(x) = \begin{cases} |x-3|, & x < 1 \\ (x-1)^4, & x = 1 \\ \sqrt{4x}, & x > 1 \end{cases}$$



Critical thinking questions:

- 9) Write a rule for the sign function $s(n)$:
 $s(n)$ is -1 when n is negative, $+1$ when n is positive, and 0 otherwise.

- 10) Write a rule for the function shown.

